# Ayush Yadav

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## SUMMARY

Software engineer, Computer Science graduate (May 2026), with a year of Python ETL and integration work in university IT. Backend in Python and TypeScript, performance code in Java. Develops daily with Claude Code, through seven custom subagents in its global config plus project level agents, and a hook that blocks deletes of protected paths and force pushes to main.

## EDUCATION & AWARDS

**Miami University, Oxford, OH | B.S. in Computer Science, Major GPA 3.65** May 2026

- Dean's List: Fall 2023, Spring 2025, Fall 2025.
- Finalist, MUCAT Design Innovation (Feb 2025): LiDAR visual assistance proposal; $2,500 prototyping grant.

## EXPERIENCE

**ITSM Data Integration Intern | Miami University | Oxford, OH** Jun 2025 – May 2026

- Built a Python ETL pipeline over five years of Oracle Analytics query logs to scope a planned Workday reporting migration to Snowflake, and found that **7 of the 50 subject areas were never queried**.
- Parsed the 1.6M queries against the repository catalog and **resolved 40.9M column reads to one physical column each**. Another 16.9M reads stayed unattributed, most of them matching several candidate columns.
- Merged 3,731 report records from Tableau's REST API and 6,743 from Workday's exports into one inventory of 10,453 rows, 21 duplicates removed, so reports can be searched in one place instead of two.
- Visualized 37 months of code compliance across 61 projects in a Tableau dashboard fed by the team's Laravel reporter, which reads TeamDynamix, GitLab and an End of Life API. Of the 61 projects, 59 are compliant and 2 at end of life.

## PROJECTS

**Applied | Gmail Sync and Application Tracking Service** 2026 – Present

*Solo | github.com/yadava5/applied | getapplied.vercel.app*

- Built a job application tracker that reads Gmail and sorts each mail into one of eight categories with 220 hand written rules. The rules were tuned on one personal inbox. On that inbox, 64 of 69 applications were filed straight from the mail.
- Isolated each user's rows in the database, not the query layer, with **all 9 tables forced under row level security** and 35 policies. A query with no WHERE clause still cannot read another user's rows.
- Rekeyed the unique email index onto the user and message id pair, closing a defect where a second account with the same message id would have failed its whole sync. **Resync is idempotent**: six passes over three mails leave three records.

**jetpack | Parallel gzip Compression Engine on JDK 25** 2026

*Solo | github.com/yadava5/jetpack-compress | jetpack-compress.vercel.app*

- Wrote a parallel gzip engine that **raised throughput from 66 to 422 MB/s on a 32 MiB synthetic corpus** by compressing one file on 10 cores, where Java's standard library uses one. The output stays a gzip file that gunzip opens.
- Compressed a 4 GiB and a 64 MiB file **inside the same 48 MiB heap**, with two blocks in flight per core, one virtual thread each.
- Vectorized Adler-32 by hand with Java's Vector API for file fingerprints in the CLI. It matches java.util.zip bit for bit and runs **2.8 times the scalar loop at 4.26 GB/s**. The JDK's own intrinsic is still faster, at 14.06 GB/s.

**Cadence | Natural Language Calendar and Tasks** 2023 – 2026

*Solo | github.com/yadava5/cadence*

- Developed a calendar and task app whose four stage parser **turns plain English into events and tasks**, reading dates, hashtags, priority and named entities. Google Calendar sync is one way, from 30 days back to 90 days ahead.
- Migrated 15 tenant isolation tests onto Testcontainers, so the suite starts its own postgres:16 in Docker and runs locally as well as in CI. **CI fails when the suite reports zero tests or any skip.**

## TECHNICAL SKILLS

**Languages & Web:** Python, Java, C++, TypeScript, SQL, HTML, CSS, React, Next.js, Tailwind CSS, Node.js, Express, FastAPI, REST APIs

**Machine Learning & AI:** PyTorch, scikit-learn, CatBoost, XGBoost, LightGBM, SetFit, ONNX, SHAP, RAG, OpenAI API, embeddings

**Data & Infrastructure:** PostgreSQL (row level security), Supabase, Snowflake, DuckDB, Parquet, pandas, SQLite, ETL, Tableau, Oracle Analytics, OpenMP, SIMD, Vector API, LangGraph, MCP, Docker, JMH, Git, GitHub Actions (CI/CD), Linux, pytest, Vitest